Oxford Prospects and Global Development Institute

Course: STEM: Maths, Physics, Computer Science and Engineering

Session Title: Intelligent Manufacturing

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Learning Objectives:
To understand what Industry 4.0 is, and the goals, drivers, and challenges behind it.
Outline:
- What is Industry 4.0 - Goals, Drivers and Challenges - Key Technologies - Industrial Examples
Main Points:

What is Industry 4.0?

Industry 4.0 is the standard term used to talk about intelligent manufacturing. This lecture discussed what businesses want from Industry 4.0, and their goals moving forward.

Industry 4.0 is a methodology and an approach, and is not a specific technology. It is about helping people work more efficiently. Alongside the industry 4.0 approach, good business practice is still necessary. Key to this lecture is cutting through the hype around industry 4.0 and helping to understand what it is and what it is not.

Industry 4.0 was originally developed by the German government in 2011 to promote automation in industry. There are several different terms for Industry 4.0 which broadly mean the same thing, for example intelligent manufacturing or smart manufacturing.

Is known as Industry 4.0 because it is supposedly the 4th industrial revolution.

The first industrial revolution (1760-1840) was to do with mechanisation – it focussed on steam and water power to produce steels/iron amongst other things, and had the first mass production.

The second industrial revolution (1871-1914) was the technological revolution. Railways meant products could be transported, and telephone networks aided communicating. This allowed more complicated products to be manufactured, for example the engine in the model T ford.

The third industrial revolution (1947-present) was the digital revolution, where the transistor was developed amongst other things. This was the first use of automation in industry, and was the beginning of industrial robots being used in manufacturing. Smartphones and mobile phones aided with this from a communications point of view.

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Industry 4.0 is now a global movement, with lots of traction. It can involve simple fixes to improve productivity. For example, at Renault, 45 minutes was saved per shop floor employee per day, simply by improving the wifi and providing the employees with smart devices.

Elements of Industry 4.0

Industry 4.0 is comprised of *connectivity* and *intelligence*. Connectivity involves getting data and getting it to where it needs to be, and intelligence involves understanding that data.

Connectivity –

Machine to Machine Connectivity (M2M) – This is a key factor of Industry 4.0, and happens automatically. The data does not need to be moved by people, instead, an algorithm finds and fetches the data. This isn't a new process, for years, machines on the shopfloor have been communicating using field buses. Field buses are discussed later, but the newer M2M process fits better into Industry 4.0.

Internet of Things (IoT) – The internet of things involves adding connectivity to everything involved in the system. Everything is connected to a network, and so there are fewer hurdles to collecting data from a system.

Smart Sensors – You need to know what your system is doing in order to be able to control your system. These sensors already exist, but smart sensors involves making the most of them.

Location Detection – Location Detection is more of a fringe technology in Industry 4.0. Understanding where things are in your factory can aid you to optimise your processes. This can help products be tracked beyond the factory and in their service life, which can help understanding of how the product is being used.

Human machine interfaces – Humans are still the most adaptable resources. Industry 4.0 isn't about removing people from the system, it is about optimising so that humans do what they are good at, and computers and robots do what they are good at.

Intelligence –

Real time data – Real time data means that there is no delay as data is gathered or sent. The information is delivered as soon as it is collected. In the past, data often had to be inputted manually which slowed processes down. In Industry 4.0, there are constant streams of data that can be accessed and used immediately.

Business Intelligence and Data Analytics – Business Intelligence means making the data streams understandable so that the information can be interpreted and used to understand what has happened. Data Analytics means that data can be used to help predict the future.

Big Data – Big Data provides techniques for managing large and unstructured (not well organised) collections of data.

Data Mining – It is key in data mining to find value in the data, for example finding patterns which can help extract insights, groups and anomalies automatically. This can help businesses to understand why things are happening.

Machine Learning – Machine learning uses algorithms to create a model which improves with experience, and learns from large data sets. This is better than previous models, where humans would create the models and then have to update them manually.

Goals, Drivers and Challenges for Industry 4.0 –

Driving Forces -

The following describes the driving forces for Industry 4.0:

Growing Competition – Every company wants an edge over its competitors – there is growing competition both nationally and internationally.

Increased Innovation Capacity and Productivity – Productivity is higher than it has ever been. There are more complicated products than there have ever been. There is a drive to improve, to reduce the design time and the time it take to take a product to market.

Expectations of Customers – Heterogenous markets, for example with Amazon - people know they can get next day delivery, why would they go anywhere else. There are shorter product life cycles, because people don't want old products.

Sustainability – Materials are more expensive and energy costs are high. It benefits companies to waste less energy, material and time.

Financial factors – A lot of Industry 4.0 focusses on smart maintenance and improvements to reduce operating costs. Improved product quality is also a factor here.
Management support – Industry 4.0 means that more data is available to allow decisions to be made in businesses.

Business Model Innovation – Industry 4.0 allows for innovation on the business model. For example, Airbnb is the biggest hotel company and Uber is the biggest taxi company – neither of these

companies owns hotels or taxis, but they have software that connects the people who have the resources with the people that need them.

Goals –

The Industry 4.0 methodology is about allowing humans to do what humans are good at, and robots to do what robots are good at. For example, a 3-5% productivity increase using an Industry 4.0 approach can free up 45-55% productivity for knowledge workers. This means the humans in the system are able to do what they are good at, and have more time to think and design, which can improve the system further.

This leads on to the goals for the Industry 4.0 approach, which are discussed below:

Optimised and Automated Production – Productivity can be increased by automating and optimising production.

Real-Time Data for Real-Time Decisions – This is a key goal for Industry 4.0. This will mean that decisions can be based on the data, as it is occurring. This means changes in the market, supply chain and on the shop floor can be reacted to faster.

Business Continuity – If all business assets are connected, then the business is able to react to problems before they develop into problems.

Better Products – Industry 4.0 will allow the introduction of better products, for example higher quality products with more features.

Better Working Conditions – The 4D processes can be automated. 4D stands for Dirty, Dull, Dangerous and Dear (another word for expensive). Industry 4.0 will also allow for incidents to be more quickly identifies and responded to.

Agility – The agility of an industry 4.0 methodology means that production can be reconfigured more quickly, allowing the business to adapt and improve with improved visibility and forecasting of demands.

Business Model Innovation – In an Industry 4.0 system information will be readily available, so this will allow innovation in the business model, and the ability to tap into new revenue sources.

Challenges –

There are several challenges for the Industry 4.0 approach, which will be discussed below.

Lack of a Digital Strategy – A strong leadership and direction is necessary, because Industry 4.0 affects all parts of the business. Therefore, a clear strategy is important.

Lack of Established Standards and Certification – In an emerging area such as this, the regulations often haven't caught up with the technology. The safety regulations can be interpreted differently, because the standards have not yet been defined and settled on.

Lack of Skills – Manufacturing skills are declining, and digital skills are increasing. It is rare to find people who have both. Universities are trying to address this by adding digital skills to manufacturing courses.

Lack of Business Cases – New technologies often aren't well understood, so it can be hard to assess the technology and give a clear business case. This means companies often won't spend the money on the new technology.

Lack of Infrastructure – For a company which is reliant on cloud services, it is crucial to have a great internet connection. This means a solid digital infrastructure is necessary.

Cybersecurity – This links into the digital infrastructure, as if you are to be reliant on cloud services, then the cybersecurity needs to be solid. The more linkages and connectivity in an Industry 4.0 approach mean there are more potential avenues for breaches.

Resistance to Change – People are often unwilling to adapt to new technologies. There can also be a distrust of the new technologies.

Key Technologies -

Internet of Things –

The Internet of Things (IoT) is often used in home applications, and there is also an industrial IoT (IIoT) – the difference between the IoT and IIoT is very small. The use of the word things is intentional, because it can be any number of things that connect – for example a smart card or a lightbulb.

An IoT can generate data, which then needs to be analysed. The data needs to be understandable in order to be used.

Machine to Machine Communication –

Machine to Machine Communication (M2M) enables direct communication and collaboration between devices.

Traditionally systems used field buses, which are rigid and hierarchical. In this kind of network, machines can only talk to the things above them in the system. They can't communicate with other sensors on the same level in their network.

M2M allows machines to communicate among each other so data can be accessed where and when you need it. These communications occur via the software and not via cables. These can have latency, so for real time controls often the old system works better. M2M is good for data analysis.

Data Storage –

In industry, lots of the data collected is not ever used. Data needs to be visible, so that it is available where and when it is required. This storage needs to be transparent, e.g. not in the form of a string of numbers, and the data needs to be understandable and usable as required. Data needs to effectively communicated and understood in order to be useful, so you are not just storing the data for the sake of storing.

Business Intelligence -

The data needs to be utilised in order to create information and make better decisions. The data is not easy to understand alone, so visuals need to be created which allow us to understand. This means that data driven decisions can be made.

Data Analytics –

Data Analytics can be used to analyse data to understand what happened, why things happen and what will happen next. Data analytics allow more insights to be extracted from the data using new techniques.

Case Studies –

Technology Maturity
High Maturity – made from off the shelf technologies
Low Maturity – not well understood, can't just go out and buy these technologies

Data and Industry 4.0

Industry 4.0 relies on data and ways to understand the data. It is a loop. Internet of Things and sensors gather the data which then goes into the system. Humans are involved in decision making and understanding. Humans can then be used to improve the processes.

Processes which generate data have more applicability to an Industry 4.0 world, whereas 3d printing and robotics are used by an Industry 4.0 world. They are being driven by the industry 4.0, as opposed to driving 4.0. The processes in Industry 4.0 allow these techniques to be used.

Industry Example – ABB

ABB is a company which manufactures medium voltage switchgear.

Previously, all their processes were done manually. No robot communicated with each other, and they wanted to automate their processes. They went to look at solutions from other countries.

Now, they have autonomous logistics through AGVs. All their products have RFID tags which mean they can find the locations of all their parts and products. At the time they were trying to improve their processes, there weren't many market ready solutions, so everything had to be tweaked just for them.

Manufacturing in Space

NASA simulate the microgravity of manufacturing in space using swimming pools, and used this to practice how automation of these processes could work in space. Now they can have in-orbit service and manufacturing, and can assemble and maintain another satellite in space. This is better than sending astronauts into space, because this is very dangerous and expensive. Communication is key for this, and these communication technologies already exist.

For the ISS, modules were initially assembled on the ground, and brought up into space. To begin with, they were added on by a person. Then, the pilot robotically did it, and now they are autonomously docked. The future is to assemble the modules in space, which can be done using the industry 4.0 approach.

Questions

How can we understand humans in the loop? Is it possible to get humans out of the loop in the short future?
From a technological perspective, it is possible to remove humans from the loop. However, it may not be economically viable. 80 percent of the time you can remove people from the system, but 20 percent of the time it would just be so complicated and expensive to design systems to remove the humans. This is because humans are just so flexible and adaptable – we can understand when we are going to break something for example. There is also a regulatory aspect – often, we need a person there to make the decisions, for example whether a flaw is suitable for use, or to make decisions about the process.

Is M2M this the same as decentralisation? Is the algorithm used the same as bitcoin?

Decentralisation is one of the aspects for M2M. We don't have to decentralise everything - the system can still be controlled the same way but the data sharing can be done via decentralisation. The bitcoin algorithm is not directly related, but those processes have been proposed to be looked at for using in manufacturing.

With increased automation, if there is an error, how do we prevent it becoming a chain reaction and influencing the whole system?

When you write code, try to design the code so you can predict these errors and decide what to do. Use the error information as part of the information that you are building the system from, and build in error handling into the system. Industry 4.0 is trying to predict when these errors are going to happen, and stop the system before these errors propagate.
Conclusions:
Industry 4.0 is a philosophy for intelligent manufacturing. It is a wide ranging approach for capturing, analysing and using data in manufacturing. It has potential benefits for many areas of businesses, from increased productivity, increased quality and new business models. There are many technologies which can be used to enable industry 4.0 which have many different applications.
Further Reading:
nottingham.ac.uk/manufacturing